

```

clc; clear;
Wp=0.4; Ws=0.55; Rp=1; As=40;
[N,Wn]=cheb1ord(Wp,Ws,Rp,As);
[b,a]=cheby1(N,Rp,Wn);
fprintf('Filter Order = %d\n',N);
fprintf('Cutoff Frequency = %.2f * pi rad/sample\n',Wn);
if all(abs(roots(a))<1)
disp('System is STABLE');
else
disp('System is UNSTABLE');
end
% ----- GRAPHS -----
figure; freqz(b,a);
% Magnitude + Phase
figure; impz(b,a); % Impulse Response
figure; grpdelay(b,a);
% Group Delay
figure; zplane(b,a); % Pole-Zero

```

